Energy Decay in a Damped Harmonic Oscillator

Lena Ortiz
Department of Physics, Northfield College

October 2026

Abstract

We study the free motion of a mass on a linear spring subject to viscous damping. We derive the closed-form solution in the underdamped regime, show that the mechanical energy decays monotonically, and compare the analytic solution with two simple numerical schemes.

1 Introduction

The damped harmonic oscillator is the simplest model of a system that loses energy to its surroundings [1]. Its equation of motion is

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{1}$$

where $m > 0$ is the mass, $c \geq 0$ the damping coefficient and $k > 0$ the spring constant. Dividing by m gives the standard form

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^{2x} = 0 \tag{2}$$

2 Solution

Substituting $x = e^{\lambda}t$ into (2) gives the characteristic equation $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, whose roots are

$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} \tag{3}$$

The behaviour depends on the sign of the discriminant:

$$regime = \begin{cases} underdamped, & \gamma < \omega_0, \\ critically & damped, \end{cases} \gamma = \omega_0, \tag{4}$$

$$overdamped, \tag{5}$$

3 Numerical check

We integrated (2) with $\omega_0 = 2$, $\gamma = 0.2$ and $x_0 = 1$ on $t \in [0, 10]$, using the explicit Euler method and the classical fourth-order Runge–Kutta method (RK4) [2]. For each step size h we:

1. integrated from t = 0 to t = 10 with 10/h steps;
2. compared every step with the exact solution (4);
3. recorded the largest absolute difference.

Table 1: Table 1: Maximum absolute error on [0, 10].

Quantity	Value	Uncertainty
a	1.00	0.02
b	2.50	0.05

References

[1] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, 3rd ed., Addison-Wesley, 2002.

[2] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I*, 2nd ed., Springer, 1993.